Project Report

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Contents

1 Entanglement Dynamics in Kicked Ising Model **3**
1.1 Defining the Hamiltonian . . . 3
1.1.1 Analysis with Thue Morse Sequence . . . 4
1.1.2 Evolution with different initial states . . . 5
1.2 Calculating Page entropy . . . 6
1.3 Note on Jordan Wigner transformations . . . 9
1.3.1 Bogoliubov transformation . . . 10
1.3.2 Unitary operator after Jordan Wigner . . . 11
1.4 Geometric picture . . . 15
1.5 Entanglement Entropy of the fermion subsystem . . . 15
1.5.1 Calculating Reduced density matrix from correlation functions . . . 17

2 Remarks **19**

1 Entanglement Dynamics in Kicked Ising Model

1.1 Defining the Hamiltonian

Consider a 1-D spin chain with interactions only along the x-axis, and a pulsed magnetic field along z-axis which acts on all spins at once.

Hamiltonian for kicked Ising model with periodic boundary conditions is given by:

$$H(t) = \sum_{j=1}^{L} \sigma_j^x \sigma_{j+1}^x + \sum_{k=-\infty}^{\infty} \delta(k - \frac{t}{T}) \sum_{j=1}^{L} \sigma_j^z \tag{1}$$

Here T is the time period after which the kicks are applied to the Ising chain.

Deriving time evolution operator for Floquet system:

The Schrödinger equation for this kicked Ising model is given by (let $\sum_{j=1}^{L} \sigma_j^x \sigma_{j+1}^x = H_0$ and $\sum_{j=1}^{L} \sigma_j^z = H_1$),

$$i\hbar \frac{\partial}{\partial t}|\psi\rangle = H_0|\psi\rangle + H_1 \delta_T(t)|\psi\rangle \tag{2}$$

Here $\delta_T(t) = \sum_{k=-\infty}^{\infty} \delta(k - \frac{t}{T})$. In the time interval between two consecutive kicks, only H_0 contributes to the unitary operator. So, therefore we define:

$$U_{after} = \exp(-iH_0 T/\hbar) \tag{3}$$

Here U_{after} is the unitary operator that acts between consecutive kicks. Now, in order to find the unitary evolution operator at the time of the kick, we integrate the Schrödinger equation over an infinetisimal time about $t = T$. This eliminates the H_0 part and we will be left with:

$$i\hbar \frac{\partial}{\partial t}|\psi\rangle = H_1 \delta_T(t)|\psi\rangle \tag{4}$$

Solving by integrating over an infinitesimal time duration ϵ, we get (and setting $\hbar = 1$):

$$i \log(\frac{|\psi(T+\epsilon)\rangle}{|\psi(T-\epsilon)\rangle}) = \int_{T-\epsilon}^{T+\epsilon} H_1 \delta_T(t) \tag{5}$$

$$|\psi(T+\epsilon)\rangle = \exp(-iH_1 T)|\psi(T-\epsilon)\rangle \tag{6}$$

This gives the time evolution operator at the time of kick to be:

$$U_{kick} = \exp(-iH_1 T) \tag{7}$$

In this study, we consider the evolution with the time period $\frac{\pi}{4}$ [3]. Hence the overall unitary operator driving the system for one time period would be:

$$U = U_{after} U_{kick} = \exp(-i\frac{\pi}{4} \sum_{j=1}^{L} \sigma_j^x \sigma_{j+1}^x) \exp(-i\frac{\pi}{4} \sum_{j=1}^{L} \sigma_j^z) \tag{8}$$

Making the Hamiltonian aperiodic by incorporating the idea from [4], where the nth kick occurring at time $t = nT$ is shifted to $t = nT + \tau_n dt$ where $\tau_n = \pm 1$, and we explore the behaviour of the system for various sequences of τ_n's in the form for example, $\tau = \{1, -1, 1, -1, -1, 1, \cdots\}$. The Dirac train in the Hamiltonian for this following case takes the form:

$$\delta_T(t) = \sum_{k=-\infty}^{\infty} \delta(k - \frac{t - \tau_n dt}{T}) \tag{9}$$

In this case, the time interval between consecutive kicks becomes $T + (\tau_{n+1} - \tau_n)dt$. So, this changes the time evolution operators in the following ways:

$$U_{after} = \exp(-i(\frac{\pi}{4} + (\tau_{n+1} - \tau_n)\mathrm{dt}) \sum_{j=1}^{L} \sigma_j^x \sigma_{j+1}^x) \tag{10}$$

$$U_{kick} = \exp(-i(\frac{\pi}{4}) \sum_{j=1}^{L} \sigma_j^z) \tag{11}$$

With this, the overall time evolution operator acting the nth time takes the form:

$$U_n = \exp[-i\frac{\pi}{4}(1 + (\tau_{n+1} - \tau_n)\frac{\mathrm{dt}}{\pi/4}) \sum_{j=1}^{L} \sigma_j^x \sigma_{j+1}^x] \exp[-i\frac{\pi}{4} \sum_{j=1}^{L} \sigma_j^z] \tag{12}$$

Since $\tau_n = \pm 1$, we define a sequence μ such that $\mu_n = (\tau_{n+1} - \tau_n)$. So, $\mu \, \epsilon \, \{2, 0, -2\}$. For the cases where $\dfrac{\mathrm{dt}}{\pi/4} \ll 1$, the evolution of the subsystem entanglement entropy with number of kicks resembles that of the evolution with $\mathrm{dt} = 0$.

Now we explore the special case where $\mathrm{dt} = \dfrac{\pi}{8}$.

$$U_{after} = \exp(-i(\frac{\pi}{4} + \mu_n \frac{\pi}{8}) \sum_{j=1}^{L} \sigma_j^x \sigma_{j+1}^x) \tag{13}$$

$$U_{after} = \begin{cases} \exp(-i\frac{\pi}{4} \sum_{j=1}^{L} \sigma_j^x \sigma_{j+1}^x) & \mu_n = 0 \\ 1 & \mu_n = -2 \\ \exp(-i\frac{\pi}{2} \sum_{j=1}^{L} \sigma_j^x \sigma_{j+1}^x) & \mu_n = 2 \end{cases} \tag{14}$$

$$\tag{15}$$

Notice the case $\mu_n = 2$. Here,

$$\exp(-i\frac{\pi}{2} \sum_{j=1}^{L} \sigma_j^x \sigma_{j+1}^x) = \prod_{j=1}^{L} \exp(-i\frac{\pi}{2} \sigma_j^x \sigma_{j+1}^x) \tag{16}$$

$$= \prod_{j=1}^{L} (-i\sigma_j^x \sigma_{j+1}^x) \tag{17}$$

$$= (-i)^L \tag{18}$$

So the entanglement properties of the system is not affected whenever $\mu_n \, \epsilon \, \{2, -2\}$.

1.1.1 Analysis with Thue Morse Sequence

Consider the Thue-Morse sequence given by $\tau_{tm} = \{1, -1, -1, 1, -1, 1, 1, -1,$
$-1, 1, 1, -1, 1, -1, -1, \cdots\}$. And $\mu_{tm} = \{-2, 0, 2, -2, 2, 0, -2, 0, 2, 0, -2, 2, -2, 0, 2, -2, 2, \cdots\}$. Consider the initial state to be the $|+\rangle_{\otimes L}$ state (eigenstate of the interaction term) where $|+\rangle = \dfrac{|1\rangle + |0\rangle}{\sqrt{2}}$ and $|1\rangle$ and $|0\rangle$ are the eigenvectors of σ^z with eigen values -1 and 1 respectively.

The state after n kicks is given by:

$$|\psi(n)\rangle = U_n U_{n-1} \cdots U_2 U_1 |\psi(0)\rangle \tag{19}$$

A nice property of the Thue Morse sequence is that it is cube-free, from this we can conclude that in the sequence $\mu \, \epsilon \, \{-2, 0, 2\}$, there appears no occurances of consecutive 0's. This results in the entanglement entropy of the subsystem oscillating between only 2 values one of which is 0 as can be seen in Figure:**??**. Taking the case where the initial state of the system is $|+\rangle_{\otimes L}$, the subsystem entanglement entropy stays at 0 for all times.

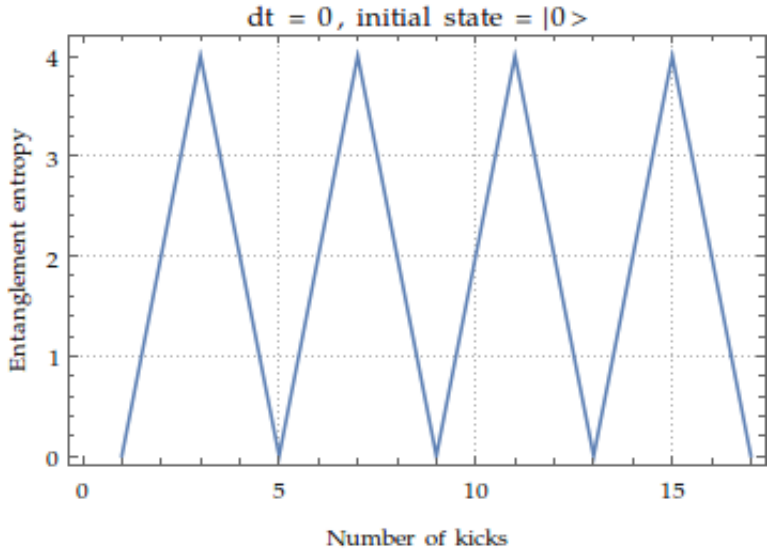


Figure 1: Starting with the initial state $|0\rangle$ in a 8-spin chain, entanglement entropy of the subsystem consisting of first 4 spins is being plotted with time for dt = 0

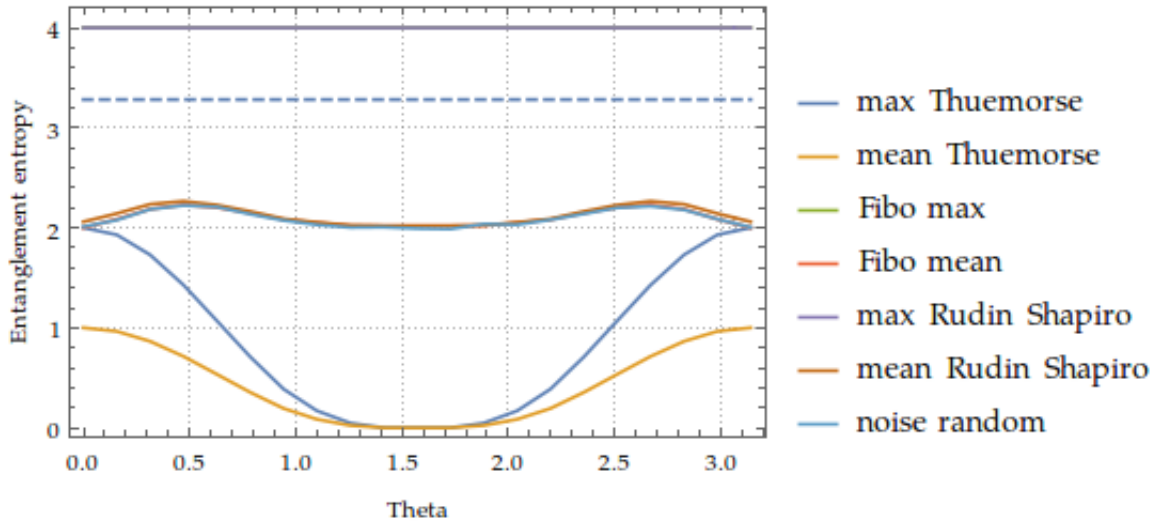


Figure 2: Plot of entanglement entropy for the aperiodic hamiltonian with different sequences for different initial states

1.1.2 Evolution with different initial states

Consider the initial state to be.

$$|\psi\rangle = \prod_{\otimes L}(\cos\frac{\theta}{2}|0\rangle + e^{i\phi}\sin\frac{\theta}{2}|1\rangle) \tag{20}$$

Since the hamiltonian does not have a σ^y term, varying ϕ will not affect the dynamics of the system. Now studying the dynamics of $|\psi(n)\rangle$ (where n is the number of kicks) for different values of θ (for 2 spin system).

For this system, the Unitary operator with periodic boundary condition would be:

$$U = \exp(-i\frac{\pi}{2}\sigma_1^x\sigma_2^x)\exp(-i\frac{\pi}{4}(\sigma_1^z+\sigma_2^z)) \tag{21}$$

$$= -i\sigma_1^x\sigma_2^x + \frac{1-i(\sigma_1^z+\sigma_2^z)}{\sqrt{2}} \tag{22}$$

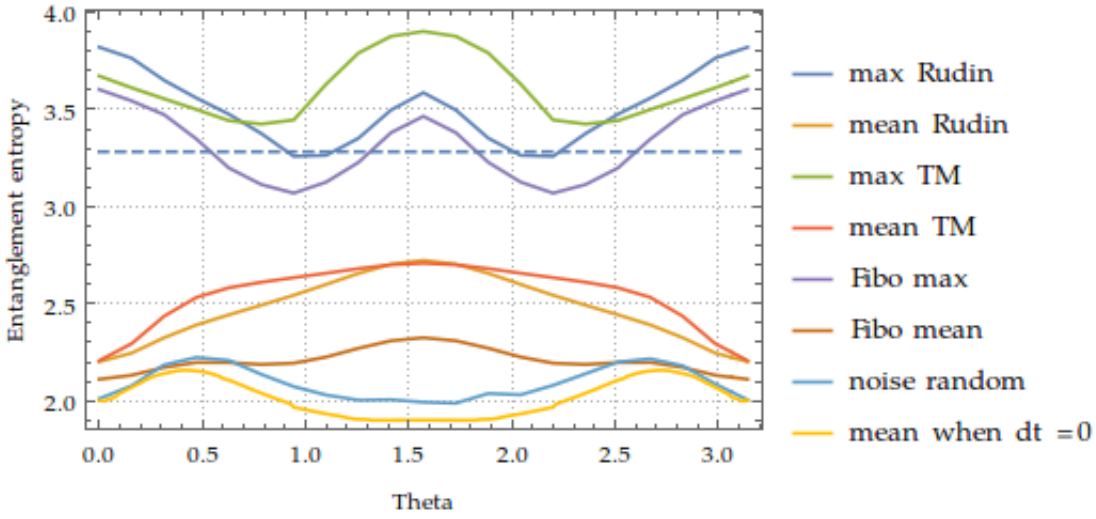


Figure 3: Plot of entanglement entropy for the aperiodic hamiltonian with sequences itself and not the first difference present in the Unitary operator for different initial states

Case 1 :
$\theta = 0$

$$|\psi(0)\rangle = |00\rangle \tag{23}$$

$$|\psi(1)\rangle = \exp(-i\frac{\pi}{4}\sigma_1^x\sigma_2^x)\exp(-i\frac{\pi}{4}\sum_{j=1}^{2}\sigma_j^z)|00\rangle \tag{24}$$

$$= \frac{1}{2\sqrt{2}}(I - i\sigma_1^x\sigma_2^x)(I - i\sigma_1^z)(I - i\sigma_2^z)|00\rangle \tag{25}$$

$$= \frac{1}{2\sqrt{2}}(1-i)^2(|00\rangle + i|11\rangle) \tag{26}$$

$$|\psi(2)\rangle = \frac{1}{2\sqrt{2}}((1-i)^4(|00\rangle + i|11\rangle) + i(1-i)^2(1+i)^2(|00\rangle + i|11\rangle) \tag{27}$$

1.2 Calculating Page entropy

Page entropy is the average entropy of a subsystem if the quantum system of the Hilbert space is in a pure random state. If the dimension of the Hilbert space is mn the average entropy of the subsystem of dimension $m \leq n$ is given by $S_{m,n} = \sum_{k=n+1}^{mn}\frac{1}{k} - \frac{m-1}{2n}$. In this case, consider an 8-spin ising chain, so the dimension of the Hilbert space is 2^8 and the subsystem with 4 spins will result in $m = 2^4$ and $n = 2^4$. So,

$$S_{16,16} = \sum_{k=17}^{256}\frac{1}{k} - \frac{15}{32} \tag{28}$$

$$= 2.274865969588287 \tag{29}$$

But this value is taken with $\log_e$ but we have calculated the entanglement entropy with $\log_2$, so, diving the above obatined value with $\log_e 2$ we get Page entropy to be:

$$S_{16,16} = \frac{2.274865969588287}{\log_e 2} = 3.281937853012085 \tag{30}$$

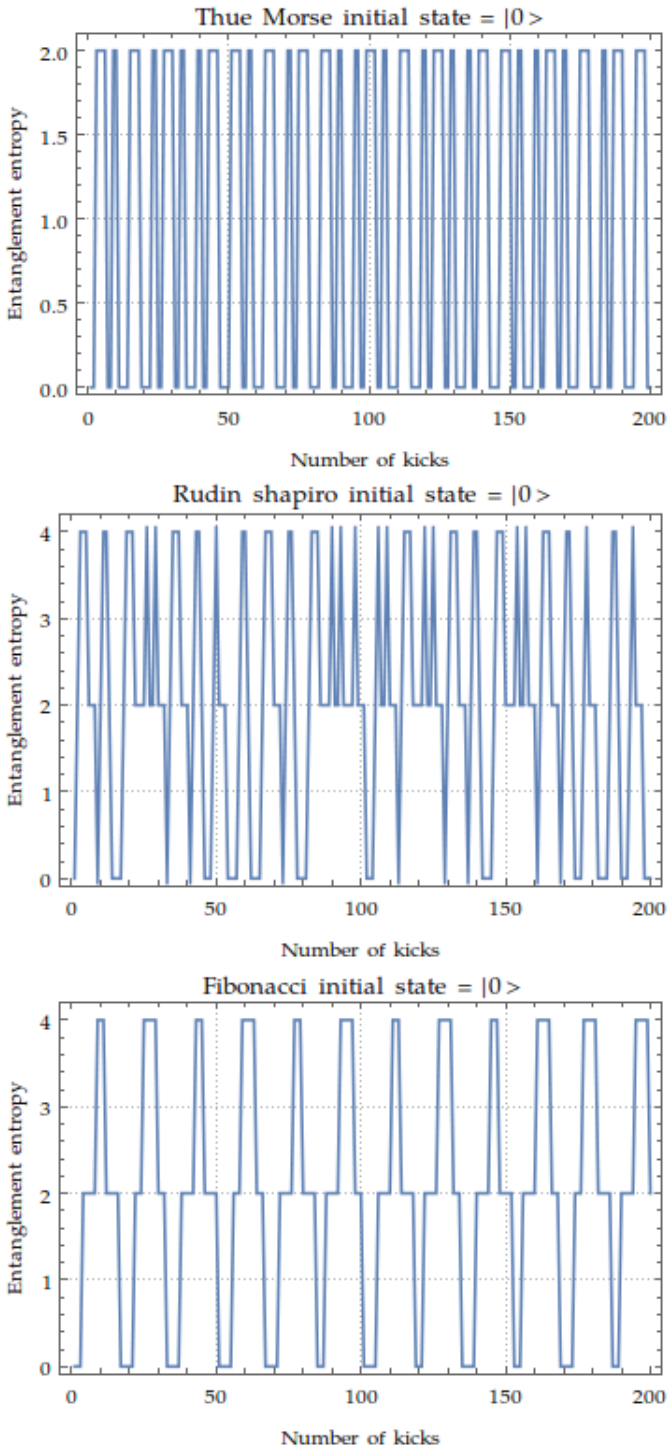


Figure 4: The Unitary operator contains the first difference of the above given sequces and the system was let to evolve with dt = pi/8

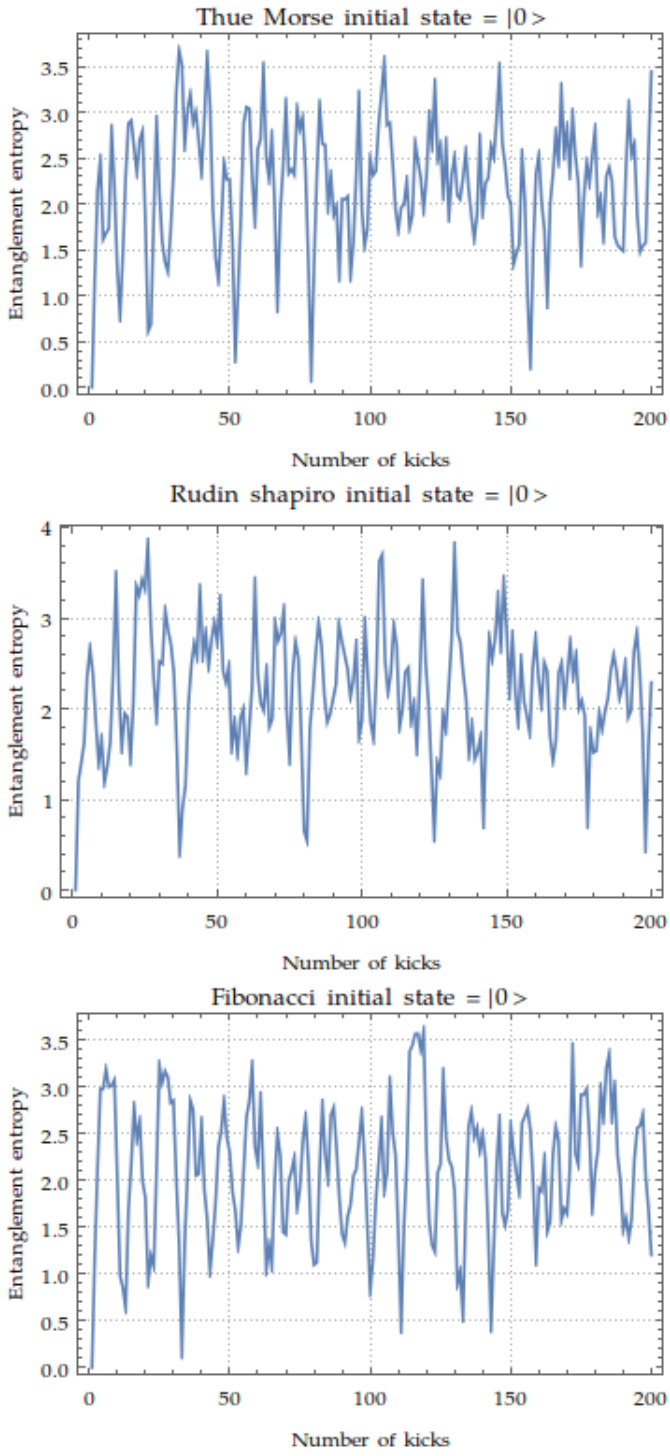


Figure 5: The Unitary operator contains the sequence itself intead of the fisrt difference of the above given sequces and the system was let to evolve with dt = pi/8

1.3 Note on Jordan Wigner transformations

Let the Hamiltonian for 1D transverse field Ising chain be given in the following form:

$$H_{IM} = -J\sum_{n=1}^{N}\sigma_n^z\sigma_{n+1}^z - h\sum_{n=1}^{N}\sigma_n^x \tag{31}$$

Consider the fermionic operator, c_j at the jth site such that it satisfies the following commutation relations:

$$\{c_i, c_j^\dagger\} = \delta_{ij} \qquad [c_i, c_j] = 0 \tag{32}$$

Introducing the Majorana fermion operators with attached string operators such that the fermion antisymmetric properties under interchange between different sites is preserved is given by:

$$\psi_n^1 = \frac{1}{\sqrt{2}}[\prod_{l=1}^{n-1}\sigma_l^x]\sigma_n^y \tag{33}$$

$$\psi_n^2 = \frac{1}{\sqrt{2}}[\prod_{l=1}^{n-1}\sigma_l^x]\sigma_n^z \tag{34}$$

Where ψ_j^1 and ψ_j^2 also satisfy :

$$\psi_j^1 = \frac{c_j + c_j^\dagger}{2} \tag{35}$$

$$\psi_j^2 = \frac{c_j^\dagger - c_j}{2i} \tag{36}$$

These Majorana fermions satistfy the relations:

$$\{\psi_i^k, \psi_j^l\} = \delta_{ij}\delta_{kl} \tag{37}$$

From the above equations, we can express the pauli matrices in terms of Majorana fermion operators:

$$\sigma_j^x = -4i\psi_j^1\psi_j^2 \tag{38}$$

$$\sigma_j^z\sigma_{j+1}^z = 4i\psi_j^1\psi_{j+1}^2 \tag{39}$$

When periodic boundary conditions are imposed, the extra term in terms of Majorana fermions becomes $\sigma_N^z\sigma_1^z = -[\prod_1^N\sigma_l^x]2i\psi_N^1\psi_1^2 \equiv (-1)\cdot\mathcal{P}\cdot 2i\psi_N^1\psi_1^2$ where $\mathcal{P} = [\prod_1^N\sigma_l^x]$ is the parity operator.

Consider the number operator given by $c_j^\dagger c_j = \dfrac{1+2i\psi_j^1\psi_j^2}{2} = \dfrac{1-\sigma_n^x}{2}$.

Writing $\sigma^x = -ie^{1\pi\sigma^x/2} = e^{1\pi(\sigma^x-1)/2}$, we can note that:

$$\mathcal{P} = \prod_1^N\sigma_l^x = (-1)^{\sum_{j=1}^N c_j^\dagger c_j} \tag{40}$$

Now rewriting the Hamiltonian in terms of ψ^1 and ψ^2, we get :

$$H_{IM} = -\frac{J}{4i}\sum_{j=1}^{N}\psi_j^1\psi_{j+1}^2 - \frac{h}{4i}\sum_{j=1}^{N}\psi_j^1\psi_j^2 \tag{41}$$

Substituting Eqns 28,29 we obtain the Hamiltonian in terms of c_j's also, assuming periodic boundary conditions, we get:

$$H_{IM} = -J\sum_{j=1}^{N}(c_j^\dagger c_{j+1} + c_j^\dagger c_{j+1}^\dagger - c_j c_{j+1} - c_j c_{j+1}^\dagger) - 2h\sum_{j=1}^{N} c_j^\dagger c_j \tag{42}$$

Fourier transform of fermionic operators:

$$c_j = \frac{1}{\sqrt{N}}\sum_k d_k e^{i\pi kj/N} \tag{43}$$

$$d_k = \frac{1}{\sqrt{N}}\sum_j c_j e^{-i\pi kj/N} \tag{44}$$

Note that, in the summation k only takes odd/even values depending on antiperiodic/periodic boundary conditions. Here, since we are assming periodic boundary conditions, k takes the following values:

$$k\epsilon\{\pm 1, \pm 3, ..., \pm(N-1)\}$$

New hamiltonian in terms of d_k's:

$$H_{IM} = -J\sum_{k\geq 0}^{\pi(N-1)/N}(d_k^\dagger d_k e^{ik} + d_k^\dagger d_{-k}^\dagger e^{ik} - d_k d_{-k} e^{-ik} - d_k d_k^\dagger e^{-ik}) - 2h\sum_{k\geq 0}^{\pi(N-1)/N} d_k^\dagger d_k \tag{45}$$

Simplified form:

$$H_{IM} = -2\sum_{k\geq 0}^{\pi(N-1)/N}\begin{bmatrix} d_k^\dagger & d_{-k}\end{bmatrix}\begin{bmatrix} h + J\cos(k) & iJ\sin(k) \\ -iJ\sin(k) & -h - J\cos(k)\end{bmatrix}\begin{bmatrix} d_k \\ d_{-k}^\dagger\end{bmatrix} \tag{46}$$

1.3.1 Bogoliubov transformation

The eigen values of $M = \begin{bmatrix} h + J\cos(k) & iJ\sin(k) \\ -iJ\sin(k) & -h - J\cos(k)\end{bmatrix}$ are $\pm|B_k|$ where:

$$|B_k| = \sqrt{J^2 + h^2 + 2Jh\cos(k)} \tag{47}$$

Let $b_k = J\sin(k), c_k = h + J\cos(k)$

The eigen vectors of M are $\frac{1}{Norm}\begin{bmatrix} b_k \\ i(c_k + \sqrt{b_k^2 + c_k^2})\end{bmatrix}$ and $\frac{1}{Norm}\begin{bmatrix} i(c_k + \sqrt{b_k^2 + c_k^2}) \\ b_k\end{bmatrix}$ where $\frac{1}{Norm}$ is the normalization factor with

$$Norm = \sqrt{2\sqrt{b_k^2 + c_k^2}(c_k + \sqrt{b_k^2 + c_k^2})} \tag{48}$$

Introducing a matrix

$$U^\dagger = \frac{1}{Norm_k}\begin{bmatrix} b_k & i(c_k + \sqrt{b_k^2 + c_k^2}) \\ i(c_k + \sqrt{b_k^2 + c_k^2}) & b_k\end{bmatrix} \tag{49}$$

and rewriting the Hamiltonian:

$$H_{IM} = -\sum_{k\geq 0}^{\pi(N-1)/N} 2\begin{bmatrix} d_k^\dagger & d_{-k}\end{bmatrix} U^\dagger \begin{bmatrix} -|B_k| & 0 \\ 0 & |B_k|\end{bmatrix} U \begin{bmatrix} d_k \\ d_{-k}^\dagger\end{bmatrix} \tag{50}$$

Now let:
$\begin{bmatrix} d_k^\dagger & d_{-k}\end{bmatrix} U^\dagger := \begin{bmatrix} a_k^\dagger & a_{-k}\end{bmatrix}$ and $U\begin{bmatrix} d_k \\ d_{-k}^\dagger\end{bmatrix} := \begin{bmatrix} a_k \\ a_{-k}^\dagger\end{bmatrix}$ we get:

$$H_{IM} = -2 \sum_{k\geq 0}^{\pi(N-1)/N} \begin{bmatrix} a_k^\dagger & a_{-k} \end{bmatrix} \begin{bmatrix} -|B_k| & 0 \\ 0 & |B_k| \end{bmatrix} \begin{bmatrix} a_k \\ a_{-k}^\dagger \end{bmatrix} \tag{51}$$

Simplifying the above equation we get:

$$H_{IM} = 2 \sum_{k\geq 0}^{\pi(N-1)/N} (|B_k| a_k^\dagger a_k - |B_k| a_{-k} a_{-k}^\dagger) \tag{52}$$

Considering the fact that a_k's follow the same commutation relations as that of d_k's the hamiltonian can be further simplified into:

$$H_{IM} = 2 \sum_{k=-\pi(N-1)/N}^{\pi(N-1)/N} |B_k| (a_k^\dagger a_k - \frac{1}{2}) \tag{53}$$

1.3.2 Unitary operator after Jordan Wigner

Task: Express eqn 24 in terms of fermionic operators:

$$H_{after} = \sum_i \sigma_i^z \sigma_{i+1}^z \tag{54}$$

Here $J = -1$ and $h = 0$. So after the fourier tramsform of fermion operators the hamiltonian becomes:

$$H_{after} = 2 \sum_{k\geq 0}^{\pi(N-1)/N} \begin{bmatrix} d_k^\dagger & d_{-k} \end{bmatrix} \begin{bmatrix} \cos k & i \sin k \\ -i \sin k & -\cos k \end{bmatrix} \begin{bmatrix} d_k \\ d_{-k}^\dagger \end{bmatrix} \tag{55}$$

Consider $M = \begin{bmatrix} \cos k & i \sin k \\ -i \sin k & -\cos k \end{bmatrix}$, it has eigen values ± 1 and eigen vectors $\begin{bmatrix} i\cos(k/2) \\ \sin(k/2) \end{bmatrix}$ and $\begin{bmatrix} \sin(k/2) \\ i\cos(k/2) \end{bmatrix}$.

Now consider the matrix formed by the eigen vectors of M, $U^\dagger = \begin{bmatrix} \sin(k/2) & i\cos(k/2) \\ i\cos(k/2) & \sin(k/2) \end{bmatrix}$ such that $UMU^\dagger = \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix}$. This makes the hamiltonian:

$$H_{after} = 2 \sum_{k\geq 0}^{\pi(N-1)/N} \begin{bmatrix} d_k^\dagger & d_{-k} \end{bmatrix} U^\dagger \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} U \begin{bmatrix} d_k \\ d_{-k}^\dagger \end{bmatrix} \tag{56}$$

$$= 2 \sum_{k\geq 0}^{\pi(N-1)/N} \begin{bmatrix} d_k^\dagger & d_{-k} \end{bmatrix} \begin{bmatrix} \sin(k/2) & i\cos(k/2) \\ i\cos(k/2) & \sin(k/2) \end{bmatrix} \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \sin(k/2) & -i\cos(k/2) \\ -i\cos(k/2) & \sin(k/2) \end{bmatrix} \begin{bmatrix} d_k \\ d_{-k}^\dagger \end{bmatrix} \tag{57}$$

$$\tag{58}$$

This gives the Bogoliubov transformed operators:

$$a_k^\dagger = d_k^\dagger \sin(k/2) + i d_{-k} \cos(k/2) \tag{59}$$

$$a_{-k} = i d_k^\dagger \cos(k/2) + d_{-k} \sin(k/2) \tag{60}$$

$$a_{-k}^\dagger = -i d_k \cos(k/2) + d_{-k}^\dagger \sin(k/2) \tag{61}$$

$$a_k = d_k \sin(k/2) - i d_{-k}^\dagger \cos(k/2) \tag{62}$$

So the Hamiltonian in the duration between consecutive kicks becomes:

$$H_{after} = 2 \sum_{k\geq 0}^{\pi(N-1)/N} \begin{bmatrix} a_k^\dagger & a_{-k} \end{bmatrix} \begin{bmatrix} -1 & 0 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} a_k \\ a_{-k}^\dagger \end{bmatrix} \tag{63}$$

$$= -2 \sum_{k=-\pi(N-1)/N}^{\pi(N-1)/N} (a_k^\dagger a_k - \frac{1}{2}) \tag{64}$$

This makes the time evolution operator acting after a kick to be:

$$U_{after} = \prod_{k>0} e^{i\frac{\pi}{4} a_k^\dagger a_k} e^{i\frac{\pi}{4} a_{-k}^\dagger a_{-k}} = \prod_{k=-\pi(N-1)/N}^{\pi(N-1)/N} e^{i\frac{\pi}{4} a_k^\dagger a_k} \tag{65}$$

Note that the $\frac{1}{2}$ term in $(a_k^\dagger a_k - \frac{1}{2})$ can be ingnored as it merely adds an overall phase to the unitary operator.

Now consider the second part of the overall Hamiltonian and call it H_{kick}. Note that this is **not** exactly the Hamiltonian at the time of the kick:

$$H_{kick} = \sum_i \sigma_i^x \tag{66}$$

In this case $J = 0$ and $h = -1$. So after the Fourier transform of the fermion operators the Hamiltonian becomes:

$$H_{kick} = 2 \sum_{k\geq 0}^{\pi(N-1)/N} \begin{bmatrix} d_k^\dagger & d_{-k} \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \begin{bmatrix} d_k \\ d_{-k}^\dagger \end{bmatrix} \tag{67}$$

The simplified Hamiltonian is:

$$H_{kick} = 2 \sum_{k=-\pi(N-1)/N}^{\pi(N-1)/N} (d_k^\dagger d_k - \frac{1}{2}) \tag{68}$$

The unitary operator that generates the time evolution at the instant of kick:

$$U_{kick} = \prod_{k>0} \exp(-i\frac{\pi}{4} \sum_i \sigma_i^x) \tag{69}$$

This, after Jordan-Wigner transformation becomes:

$$U_{kick} = \prod_{k>0} e^{-i\frac{\pi}{4} d_k^\dagger d_k} e^{-i\frac{\pi}{4} d_{-k}^\dagger d_{-k}} = \prod_{k=-\pi(N-1)/N}^{\pi(N-1)/N} e^{-i\frac{\pi}{4} d_k^\dagger d_k} \tag{70}$$

Where d_k's are the same Fourier transformed fermion operators as defined before. The combined time-evolution is given by:

$$U = U_{after} U_{kick} = \prod_{k=-\pi(N-1)/N}^{\pi(N-1)/N} e^{i\frac{\pi}{4} a_k^\dagger a_k} \prod_{l=-\pi(N-1)/N}^{\pi(N-1)/N} e^{-i\frac{\pi}{4} d_l^\dagger d_l} \tag{71}$$

$$= \prod_{k=-\pi(N-1)/N}^{\pi(N-1)/N} e^{i\frac{\pi}{4} a_k^\dagger a_k} e^{-i\frac{\pi}{4} d_k^\dagger d_k} \tag{72}$$

A useful simplification:

$$e^{i\frac{\pi}{4}a_k^\dagger a_k} = 1 + a_k^\dagger a_k(e^{i\frac{\pi}{4}} - 1) = 1 + a_k^\dagger a_k(\frac{1+i}{\sqrt{2}} - 1) \tag{73}$$

Implimenting this simplification and defining $U = \prod_{k=-\pi(N-1)/N}^{\pi(N-1)/N} U_k$ we get:

$$U_k = (1 + a_k^\dagger a_k(\frac{1+i}{\sqrt{2}} - 1))(1 + d_k^\dagger d_k(\frac{1-i}{\sqrt{2}} - 1)) \tag{74}$$

Further simplification yeilds:

$$\begin{aligned} a_k^\dagger a_k &= (d_k^\dagger sin(k/2) + id_{-k}\cos(k/2))(d_k \sin(k/2) - id_{-k}^\dagger \cos(k/2)) \\ &= d_k^\dagger d_k \sin^2(k/2) - id_k^\dagger d_{-k}^\dagger cos(k/2)\sin(k/2) + d_{-k}d_{-k}^\dagger \cos^2(k/2) + id_{-k}d_k \sin(k/2)\cos(k/2) \end{aligned}$$

$$\begin{aligned} a_{-k}^\dagger a_{-k} &= (-id_k \cos(k/2) + d_{-k}^\dagger \sin(k/2))(id_k^\dagger \cos(k/2) + d_{-k}\sin(k/2)) \\ &= d_k d_k^\dagger \cos^2(k/2) - id_k d_{-k}\sin(k/2)\cos(k/2) + id_{-k}^\dagger d_k^\dagger \sin(k/2)\cos(k/2) + d_{-k}^\dagger d_{-k}\sin^2(k/2) \end{aligned}$$

$$\begin{aligned} a_k^\dagger a_k a_{-k}^\dagger a_{-k} &= -d_k^\dagger d_k d_{-k}^\dagger d_{-k}\sin^2(k/2)\cos k + d_{-k}d_{-k}^\dagger d_k d_k^\dagger \cos^2(k/2)\cos k - \\ &\quad \frac{i}{2}(d_{-k}^\dagger d_k^\dagger + d_{-k}d_k)\sin(k)\cos(k) \\ d_k^\dagger d_k d_{-k}^\dagger d_{-k} &= (1 - d_k d_k^\dagger)(1 - d_{-k}d_{-k}^\dagger) = 1 + d_k d_k^\dagger d_{-k}d_{-k}^\dagger - (d_k d_k^\dagger + d_{-k}d_{-k}^\dagger) \\ a_k^\dagger a_k a_{-k}^\dagger a_{-k} &= d_k d_k^\dagger d_{-k}d_{-k}^\dagger \cos^2 k \end{aligned}$$

$$U = 1 + (\frac{1+i}{\sqrt{2}} - 1)((\cos k + 1)(d_k^\dagger d_k + d_{-k}^\dagger d_{-k}) + \sin k(d_k d_{-k} + d_{-k}^\dagger d_k^\dagger)) \tag{75}$$

$$+ (\frac{1+i}{\sqrt{2}} - 1)^2(d_{-k}d_k^\dagger + 2d_k^\dagger d_k d_{-k}^\dagger d_{-k}(\cos k + 1)) \tag{76}$$

Note that particle number is conserved upon the action of this unitary operator. Consider the all spin up state to be $||0\rangle\rangle$. In the fermionic language this refers to state in which all the fermions occupy the lower energy level such that $c|0\rangle = 0$ and $c^\dagger|0\rangle = |1\rangle$. So, this all spin up state will be the same thing in the fermionic basis. But it is different in the basis of d_k operators.

The ground state is usually taken to be the all spin-up state. This state in the basis of d_k's turns out to be the BCS ground state.

$$|\psi\rangle_{BCS} = \prod_{k>0} \frac{1}{\sqrt{2(\cos k + 1)}}(\sin k + i(\cos k + 1)d_k^\dagger d_{-k}^\dagger)||0\rangle\rangle \tag{77}$$

Considering $\frac{\sin k}{\sqrt{2(\cos k+1)}} = \cos\theta_k$, and $\frac{\cos k + 1}{\sqrt{2(\cos k+1)}} = \sin\theta_k$ we get :

$$|\psi\rangle_{BCS} = \prod_{k>0}(\cos\theta_k + i\sin\theta_k d_k^\dagger d_{-k}^\dagger)||0\rangle\rangle \tag{78}$$

The time evolved state after n kicks looks like:

$$|\psi(n)\rangle = U^n|\psi(0)\rangle \tag{79}$$

Unitary operator for kicked Ising chain with noise incorporated:

$$U_n = \prod_k U_k = \prod_k (1 + a_k^\dagger a_k (\frac{1+i}{\sqrt{2}} e^{i(\tau_{n+1}-\tau_n)\mathrm{dt}} - 1))(1 + a_{-k}^\dagger a_{-k} (\frac{1+i}{\sqrt{2}} e^{i(\tau_{n+1}-\tau_n)\mathrm{dt}} - 1)) \tag{80}$$

$$* (1 + d_k^\dagger d_k (\frac{1-i}{\sqrt{2}} - 1))(1 + d_{-k}^\dagger d_{-k} (\frac{1-i}{\sqrt{2}} - 1)) \tag{81}$$

So clearly this system is integrable.

This exercise was being performed for the interaction term $\sigma_i^z \sigma_{i+1}^z$, similarly, for the case in which interaction is along x and magnetic field along z, the Hamiltonian in fermion momentum space turns out to be:

$$H_{kick} = 2 \sum_k [d_k^\dagger d_k - d_{-k}^\dagger d_{-k}] \tag{82}$$

$$H_{after} = -2 \sum_k (\cos k [d_k^\dagger d_k - d_{-k}^\dagger d_{-k}] - \sin k [d_{-k} d_k + d_k^\dagger d_{-k}^\dagger]) \tag{83}$$

Now let us define a new space whose basis states are $|0\rangle$ (which is the no fermion state in the momentum basis) and $d_k^\dagger d_{-k}^\dagger |0\rangle$. Hamiltonian takes this simple form in this basis:

$$H_{kick} = -2 \sum_k \sigma_k^z \tag{84}$$

$$H_{after} = 2 \sum_k (\cos k \sigma_k^z - \sin k \sigma_k^x) \tag{85}$$

The initial state is taken to be the ground state of the interaction Hamiltonian. In this newly defined basis, it takes the following form:

$$|\psi(0)\rangle_k = v_k(0)|0\rangle + u_k(0)|1\rangle \tag{86}$$

Where $v_k(0) = \sin(k/2)$ and $u_k(0) = \cos(k/2)$

Now the unitary evolution of this initial state is:

$$i \frac{d}{dt} \begin{bmatrix} v_k(t) \\ u_k(t) \end{bmatrix} = H_k \begin{bmatrix} v_k(t) \\ u_k(t) \end{bmatrix} \tag{87}$$

Now since H_k is in the form of 2 x 2 matrices, the numerical computation of the evolution of the states becomes exponentially faster.

Lets us see what the Floquet operator looks like in this case, for the particular case where the time period is $\frac{\pi}{4}$:

$$U_{kick} = \exp(-i \frac{\pi}{4} H_{kick}) \tag{88}$$

$$= \exp(i \frac{\pi}{2} \sum_k \sigma_k^z) \tag{89}$$

$$= i \prod_{\otimes k} \sigma_k^z \tag{90}$$

$$U_{after} = \exp(-i \frac{\pi}{4} H_{after}) \tag{91}$$

$$= \exp(-i \frac{\pi}{2} \sum_k (\cos k \sigma_k^z - \sin k \sigma_k^x)) \tag{92}$$

$$= -i \prod_{\otimes k} (\cos k \sigma_k^z - \sin k \sigma_k^x) \tag{93}$$

In the presence of no noise, the Floquet operator takes the simple form:

$$U = -\prod_{\otimes k}(\cos k - \sin k \sigma_k^x \sigma_k^z) \tag{94}$$

Note that this simple expression for the Floquet operator occurs only for the case when the period of the kicks happens to be $\pi/4$ which is a special case where the system sees symmetries.

In the presence of noise, the unitary operator for the kth mode turns out to be (with ϵ being 0 or 1 with a probability of 1/2) and U_j the unitary operator acting after j periods [2]:

$$U_{kick} = (i\sigma_k^z)^\epsilon \tag{95}$$
$$U_{after} = -i(\cos k \sigma_k^z - \sin k \sigma_k^x) \tag{96}$$

$$U_j = \begin{bmatrix} \cos k & -\sin k \\ -\sin k & -\cos k \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & (-1)^{\epsilon_j} \end{bmatrix} = \begin{bmatrix} \cos k & -(-1)^{\epsilon_j} \sin k \\ -\sin k & -(-1)^{\epsilon_j} \cos k \end{bmatrix} \tag{97}$$

$$U_j U_l = \begin{bmatrix} \cos k & -(-1)^{\epsilon_j} \sin k \\ -\sin k & -(-1)^{\epsilon_j} \cos k \end{bmatrix} \begin{bmatrix} \cos k & -(-1)^{\epsilon_l} \sin k \\ -\sin k & -(-1)^{\epsilon_l} \cos k \end{bmatrix} \tag{98}$$

$$= \begin{bmatrix} \cos^2 k + (-1)^{\epsilon_j} \sin^2 k & \cos k \sin k(-(-1)^{\epsilon_j} + (-1)^{\epsilon_j + \epsilon_l}) \\ \sin k \cos k(-1 + (-1)^{\epsilon_j}) & (-1)^{\epsilon_l} \sin^2 k + (-1)^{\epsilon_j + \epsilon_l} \cos^2 k \end{bmatrix} \tag{99}$$

1.4 Visualizing k-mode evolution on the Bloch sphere

Now that we have $\begin{bmatrix} v_k(t) \\ u_k(t) \end{bmatrix}$ for any t, we would want to visualize this time evolution on the Bloch sphere.

Consider a single k-mode:

$$\begin{bmatrix} v_k(t) \\ u_k(t) \end{bmatrix} = e^{i\beta} \cos[\frac{\theta}{2}]|0\rangle + e^{i\alpha} \sin[\frac{\theta}{2}]|1\rangle \tag{100}$$

$$= e^{i\beta}(\cos[\frac{\theta}{2}]|0\rangle + e^{i(\alpha-\beta)} \sin[\frac{\theta}{2}]|1\rangle) \tag{101}$$

Now we need to find out θ and $(\alpha - \beta)$. This can be found out from the following constraints.

$$|v_k|^2 + |u_k|^2 = 1 \tag{102}$$
$$\theta = 2\cos^{-1}(|v_k|) \tag{103}$$
$$e^{i\beta} = v_k / \cos(\theta/2) \tag{104}$$
$$e^{i(\alpha-\beta)} = \frac{u_k}{v_k} \cot(\theta/2) \tag{105}$$

With the parameters $\theta$and$(\alpha-\beta)$ know, we can now plot the k-mode evolution on the Bloch sphere. Notice that in this special case where the time period is taken to be $\pi/4$ the action of the Floquet operator does not bring about any change to the relative phase i.e., $(\alpha - \beta)$. We start off with the initial state whose parameters θ_0 and $\phi_0 = (\alpha_0 - \beta_0)$ are known.

1.5 Entanglement Entropy of the fermion subsystem

Finding the entanglement entropy of the subsystem from the reduced density matrix [6]. The reduced density matrix for this fermionic system can be obtained in the following manner [5] by calculating the correlation matrix [1] and relating it to the reduced density matrix of the block.

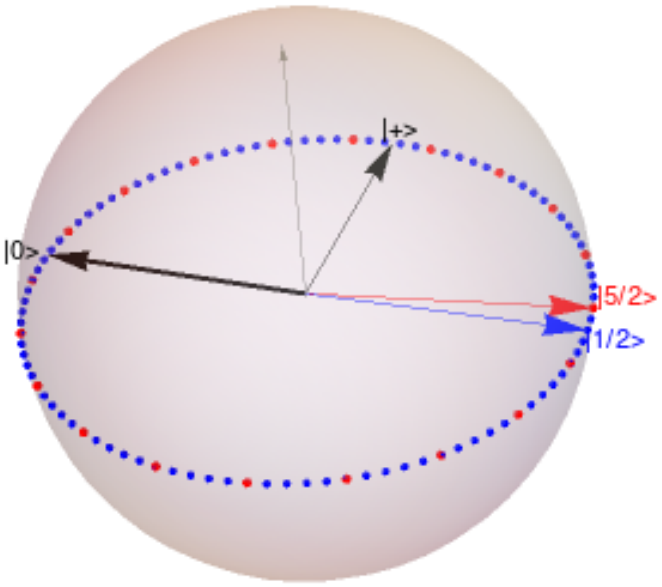


Figure 6: k-mode evolution, when $T = \pi/4$. The colored arrow shows the initial states for two modes $k = \pi/100$ and $k = 5\pi/100$

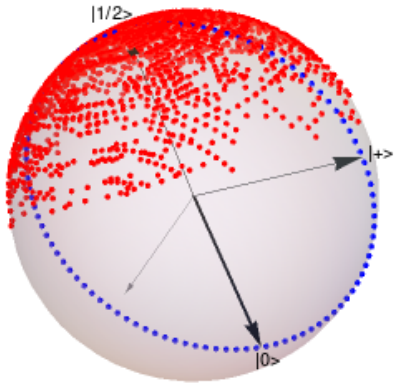


Figure 7: k-mode evolution, when $T = \pi/4 + 0.01$ shown in red for $k = \pi/100$ and the blue points correspond to $T = \pi/4$

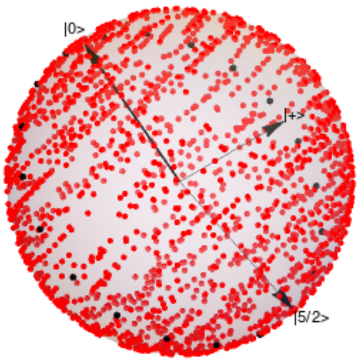


Figure 8: k-mode evolution, when $T = \pi/4 + 0.01$ shown in red for $k = 5\pi/100$ and the blue points correspond to $T = \pi/4$

1.5.1 Calculating Reduced density matrix from correlation functions

Any density matrix can be expressed as:

$$\rho = \frac{1}{\mathcal{Z}} \exp(-\beta \mathcal{H}) \tag{106}$$

Where $\mathcal{H}$ is a Hermitian matrix, β is a constant and $\mathcal{Z}$ is the normalization factor (partition function). This way of expression can be justified by noting that ρ is always positive definite and $Tr[\rho] = 1$, and Eq.(106) satisfies both these conditions.

Any general Hermititan operator for a system comprising of N 2-state particles can be written in terms of Pauli matrices and Identity operator as:

$$\mathcal{H} = \prod_{\otimes k} (\vec{B_k} \cdot \vec{\sigma} + \mu_k \mathcal{I}) \tag{107}$$

In the case of free fermions hopping between lattices the Hamiltonian has the form:

$$\hat{H} = -\sum_{n,m} \hat{t}_{nm} c_n^\dagger c_m \tag{108}$$

The reduced density matrix of a subsystem with M sites formed by the combination of eigenvectors of the above Hamiltonian can be expressed in the following form:

$$\rho = \frac{1}{\mathcal{Z}} \exp(-\sum_{i,j} H_{ij} c_i^\dagger c_j) \tag{109}$$

Let us denote $\mathcal{H} = -\sum_{i,j} H_{ij} c_i^\dagger c_j$.

Diagonalizing $\mathcal{H}$ by exploiting its eigenvectors ϕ_k and eigenvalues ϵ_k such that

$$c_i = \sum_k \phi_k(i) a_k \tag{110}$$

Now ρ becomes

$$\rho = \frac{1}{\mathcal{Z}} \exp(-\sum_{k=1}^{M} \epsilon_k a_k^\dagger a_k) \tag{111}$$

Now consider the correlation matrix given by:

$$\hat{C}_{mn} = \langle c_m^\dagger c_n \rangle \tag{112}$$

By exploiting the properties of free fermion system, we can note that:

$$\langle c_n^\dagger c_m^\dagger c_k c_l \rangle = \langle c_n^\dagger c_l \rangle \langle c_m^\dagger c_k \rangle - \langle c_n^\dagger c_k \rangle \langle c_m^\dagger c_l \rangle \tag{113}$$

We know that:

$$\langle c_m^\dagger c_n \rangle = Tr[\rho c_m^\dagger c_n] \tag{114}$$

Now the problem is to find the reduced density matrix using the correlation functions. From Wick's theorem, ρ can be decribed in terms of fermions as follows:

$$\rho = K e^{-\sum_{i,j} H_{ij} c_i^\dagger c_j} \tag{115}$$

Diagonalizng the H matrix with its eigen values being ϵ_k with eigen functions being $\phi_k(i)$. The diagonalizaiton process is done in the following way:

$$c_i = \sum_k \phi_k(i) a_k \tag{116}$$

With this the density matrix now becomes:

$$\rho = K e^{-\sum_k \epsilon_k a_k^\dagger a_k} \tag{117}$$

Finally the eigen values χ_k of $\hat{C}$ and ϵ_k of H are related by:

$$\chi_k = (e^{\epsilon_k} + 1)^{-1} \tag{118}$$

In matrix form:

$$H^T = \ln[(1 - C)/C] \tag{119}$$

Therefore ρ can be comlpetely determined once $\hat{C}$ is known.

Now consider the case of kicked Ising chain. Correlation matrix in this case after Fourier transforming the fermion operators is given by:

$$D_{kl} = \langle d_k^\dagger d_l \rangle = \frac{1}{N} \langle \sum_m e^{ikm} c_m^\dagger \sum_n e^{-iln} c_n \rangle \tag{120}$$

$$D_{kl} = \frac{1}{N} \sum_{m,n} e^{i\pi(mk-nl)/N} C_{mn} \tag{121}$$

$$C_{mn} = \frac{1}{N} \sum_{k,l} e^{-i\pi(mk-nl)/N} D_{kl} \tag{122}$$

Given that the unitary operator is given by 80 the state is evolved in the d_k basis. The correlation matrix $\hat{C}$ can be calculated by taking the Fourier transform of the correlation matrix in d_k basis. Let the initial state be defined to be $|\psi(0)\rangle$ the state after N kicks is given by $U^N|\psi(0)\rangle$. The correlation matrix after N kicks happens to be :

$$D_{mn}(N) = \frac{1}{N} \langle \psi(0)| U^{\dagger N} d_m^\dagger d_n U^N |\psi(0)\rangle \tag{123}$$

$$C_{mn}(N) = \frac{1}{N} \sum_{k,l} e^{-i\pi(mk-nl)/N} \langle \psi(0)| U^{\dagger N} d_k^\dagger d_l U^N |\psi(0)\rangle \tag{124}$$

Another correlation matrix, F is defined as:

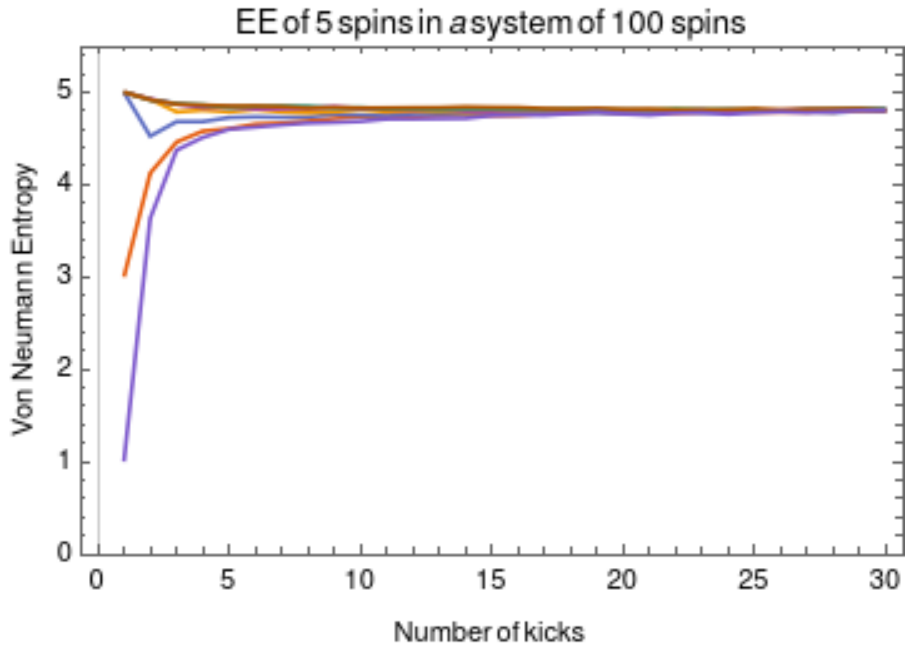


Figure 9

$$F_{mn}(N) = \frac{1}{N}\sum_{k,l} e^{-i\pi(mk+nl)/N}\langle\psi(0)|U^{\dagger N}d_k^\dagger d_l^\dagger U^N|\psi(0)\rangle \tag{125}$$

Taking into account both these correlation matrices, the $2l \times 2l$ matrix $\mathcal{C}[2l]$ is formed:

$$\mathcal{C} = \begin{bmatrix} I - C & F \\ F^\dagger & C \end{bmatrix} \tag{126}$$

So this was done for the reduced density matrix in the fermion space. How can we correlate to the reduced density matrix in the spin space?

Claim : Von Neumann entropy of the eigen values of $\mathcal{C}$ is the same as the Von Neumann entropy of the reduced density matrix of the subsystem of length l. With this, the correlation matrix can be easily calculated and we can obtain the entanglement entropy.

2 Remarks

- Eigen value spectra of the reduced density matrix needs to be calculated in order to look for signs of quantum chaos and thermalization
- Aperiodicity along the lines of Thue Morse, Fibonacci sequences also shows interesting entanglement behaviour.

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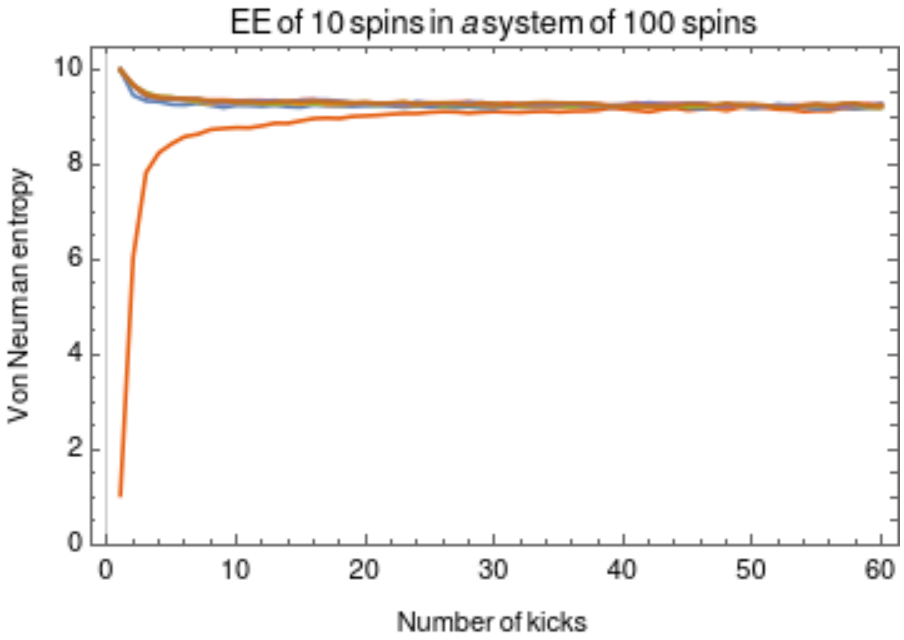


Figure 10

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